INCH-POUND

MIL-R-23417/20A
25 April 1991

SUPERSEDING
MIL-R-23417/20
22 September 1967

MILITARY SPECIFICATION SHEET

RESOLVER, ELECTRICAL, AC
TYPE 23R58W4A

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Electrical Resolvers described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) cited in the solicitation: MIL-R-23417.

TABLE I. Requirements

Requirements	Value	Unit	Tolerance
Excitation winding	Stator		
Rated frequency	400	Hz	±1%
Input voltage:			
Maximum	100	volts (rms)	±1%
Test	50	volts (rms)	±1%
Minimum	1.0	volts (rms)	±1%
Primary current	8.5	milliamps	nominal
Primary power	0.047	watts	nominal
D.C. resistance:			
Rotor	45	ohms	±15%
Stator	205	ohms	±15%
Compensator	53	ohms	±15%
Impedance:			
Zro	87+j588	ohms	R±15%+jx±10%
Zso	645+j5790	ohms	R±15%+jx±10%
Zco	82+j371	ohms	R±15%+jx+10%
Zrs	68+j55	ohms	nominal
Transformation ratio:			
Output/input	0.298		±1%
Rotor/compensator	0.825		±0.1%

AMSC - N/A FSC 5990

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TABLE I. Requirements - Continued.

Requirements	Value	Unit	Tolerance
T.R. variation (R/C) due to voltage	0.05	percent	maximum
Equality of transformation ratios:			
Output/input	0.08	percent spread	maximum
Rotor/compensator	0.1		maximum
Phase shift:			
Output/input	1.9	degrees	±0.5
Rotor/compensator	1.9	degrees	±0.5
Phase shift variation (R/C) due to voltage	±0.5	degrees	maximum
Interaxis error	±3.0	minutes	maximum
Functional error	±0.05	percent	maximum
Null voltage:			
Fundamental	10	mV(rms) output	maximum
Total	15	mV(rms) output	maximum
Shift of resolver zero:			
With voltage	±1.0	minutes	maximum
With frequency	±0.5	minutes	maximum
Harmonic distortion	0.1	percent	maximum
Temperature rise	20	degrees C	maximum
Mechanical breakaway torque:	0.2	oz-in.	maximum
@ -55°C	0.4	oz-in.	maximum
Radial play	0.0010	inches	maximum
End play	0.0030	inches	±0.0020
Resonant frequency	125	kHz	minimum
Low corner frequency	13.1	Hz	maximum

NOTE:

1. A voltage value of 1 to 100 volts shall be entered in the "volts" designation space on the identification plate.

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TABLE II. Part identifying number variant characteristics.

Part Identifying No. 1/	A ±.010	B ±.010	G Maximum	Shaft Style	Terminal End	U ±.0000 -.0005
M23417/20-01A	.676	.724	3.600	Spline	Terminal Board	.2405

1/ Part identifying number M23417/20-01A shall be the NATO standard.

1/ The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation.

TABLE III. Part identifying number cross-reference. 1/

SUPERSEDED PART NUMBERS	NEW PART IDENTIFYING NUMBERS
MIL-R-23417/20	
M23417/20-01	M23417/20-01A

1/ All line items shown in Table III refer to equivalent and interchangeable electrical resolvers of the same type designation modification. Part number changes do not affect form, fit or function of the electrical resolvers listed therein.

Custodians:
Army - AR
Navy - AS
Air Force - 85

Review activities:
Navy - EC
Air Force - 99

Preparing activity:
Army - AR

(Project 5990-0387)

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